Lab-6 Ex3 - Cubic Polynomial Evaluation

Given a cubic (degree 3) polynomial $f(x)$ in the form of

$$f(x) = ax^3 + bx^2 + cx + d$$

where a, b, c, and d are real numbers.

Write a C program to obtain from the user the four double-type coefficients a, b, c, and d. Then evaluate and print the values of the polynomial $f(x)$ with 2 decimal places for integer values x from `-6` to `4` inclusively.

For example, for x = `-6`,

$$f(-6) = a(-6)^3 + b(-6)^2 + c(-6) + d$$

Since this is clearly a finite repetition (x is varied from `-6` to `4`), try to use your newly learned `for` loop to finish the exercise.

> Note: How do you evaluate the value of `x` to the power of 3?

User-input numbers are ***bold and italic*** in the following sample run.

Sample run #1:

10 20 30 40

```
f(-6) = -1580.00

f(-5) = -860.00

f(-4) = -400.00

f(-3) = -140.00

f(-2) = -20.00

f(-1) = 20.00

f(0) = 40.00

f(1) = 100.00

f(2) = 260.00

f(3) = 580.00

f(4) = 1120.00
```